AMD DEVELOPER HACKATHON · ACT II · UNICORN TRACK · JULY 2026

aionVIS

VISION MODELS, ON DEMAND

Stop labeling data. Start describing it.

One plain English sentence becomes a trained, deployable detection model.
It generates, labels and verifies its own data. **Zero human annotation.**

Built on a single AMD MI300X | SDXL / FLUX | SAM 3 / YOLOE | YOLO · RT-DETR · RF-DETR | Gemma via vLLM | PyTorch on ROCm

Every vision model is built on a tax nobody can avoid.

COLLECT

01 **Months before a model even exists**

Thousands of on-site images, gathered before a single model can train.

LABEL

02 **$0.02 to $0.08 for every single box**

Drawn by hand at $4 to $12 an hour. A 100,000 image set runs $50,000 and up.

PAY AGAIN

03 **Change anything, and do it all over**

New camera, new product, new site. Collect, label and pay again. Data you rent, never own.

> Data labeling is a **market worth billions that exists only because models cannot feed themselves.**

Sources: LabelYourData, BasicAI, GigaBPO data annotation pricing guides (2025); industry wage surveys.

02 THE SOLUTION

One sentence in. A deployable model out. Five agents. Zero humans.

Prompt	Synthesis	Vision	Critic	MLOps
Gemma · vLLM	SDXL / FLUX	SAM 3 / YOLOE	Vision Language Model	YOLO on ROCm
Turns your sentence into hundreds of varied scenes	Renders each scene as a photorealistic image	Labels every object, any noun works	Checks and rejects bad labels. The trust layer	Trains and exports a .pt / ONNX model

On the MI300X the whole swarm stays resident, so synthesis, vision and critic run as **one parallel swarm on a single card**.

> A vision-language Critic re-checks every crop against its label and rejects the ones that fail. **Synthetic data is easy to make and hard to trust, so we ship the trust.**

194 / 244
labels the Critic rejected before training

0.85
mAP50 from 48 synthetic images, zero human labels

<$0.01
LLM cost per run, minutes of GPU time

Measured in the repo (run_0010 → model_0006): YOLOv10n, mAP50 0.85, precision 1.0, 48 images, 60 epochs. The swarm also labels your own uploaded images (BYOD).

We give eyes to agents. On demand.

One headless API, two kinds of caller: a person in the console, or an autonomous agent (drone, robot, camera) asking for a model for its exact task. Both get trained, verified weights in minutes.

A SENTENCE IN, A MODEL OUT

You describe it, or an agent asks

Send a sentence and a class list, get trained weights ready to run. No annotation, no data collection.

THE GAP IT CLOSES

General sight is too slow

Open-vocab models (SAM, OWL-ViT) see anything but are too heavy for a drone. Fixed detectors are fast but blind to new tasks. We generate the fast specialist on demand.

EXAMPLE · FARM DRONE

"Show me rotten potatoes"

A drone hits trouble in a field, asks the API for a blight detector, and runs it onboard minutes later. No connection home.

68 MB
model size

47 FPS
onboard

~97%
accuracy

$52B
AI agents market by 2030, up from $7.8B in 2025

$48.5B
AI powered drones market by 2030, driven by onboard perception

$2.9T
yearly US value from AI agents and robots by 2030 (McKinsey)

Sources: MarketsandMarkets (AI agents, AI drones); McKinsey Global Institute; Nature / Scientific Reports (UAV potato blight detection ~97%, edge models 47 FPS at 68 MB); robot foundation model surveys (open vocabulary latency trade-off).

The whole factory fits in one GPU.

AMD INSTINCT MI300X

192 GB HBM3

Diffusion, segmentation, training and the LLM all stay resident at once. No swapping, so the stages **overlap as one parallel swarm**.

- SDXL and FLUX synthesis, about **15 GB**
- SAM 3 and YOLOE vision, about **6 GB**
- YOLO trainer plus Gemma on vLLM, **resident together**
- Synthesis, vision and critic **stream in parallel**, no reloads

SINGLE NVIDIA H100

~~80 GB~~

The stack will not co-reside. Swap models per stage (slow) or split across boxes (costly). No streaming, no single-box factory.

- MI300X packs **2.4 times** the memory of an H100
- Next generation MI355X reaches **288 GB** HBM3E
- Swapping models per stage means **no parallel swarm**

DELIBERATE ORCHESTRATION

PyTorch on **ROCm** end to end, live `amd-smi` telemetry, and a run queue so **one MI300X serves a whole team.**

$4.3B

AMD data center GPU revenue in Q3 2025, up **22%** from a year earlier. Instinct is scaling.

A whole data-to-model factory, resident and running **in parallel** on one card. This is the **MI300X memory advantage** put to work.

Sources: AMD Q3 2025 results (data center GPU revenue); MI300X 192 GB HBM3 and MI355X 288 GB HBM3E specifications; memory footprints from this project's runs.

The industry stopped collecting data. It started generating it.

aionVIS is not early. It rides the biggest shift in how AI gets built: the move from gathering real data to generating it.

> The tailwind is already here. Not a forecast we are betting on, a shift already underway.

THE CROSSOVER

Synthetic passes real by 2030

Gartner: synthetic overtakes real in AI training by 2030. Already 60% of dev data in 2024, heading to 75% by 2026.

THE FRONTIER ALREADY DOES IT

Proven at the top

NVIDIA, Tesla, Waymo and others already train on generated data. The method is settled. The missing piece is trusting the labels, which is what we ship.

REAL DATA RUNS OUT

You cannot collect your way there

Usable human data runs dry between 2026 and 2032. And the rarest cases (defects, edge scenes, objects in orbit) never had a dataset at all.

2030
synthetic data overtakes real in AI training (Gartner)

~35% / yr
growth of the synthetic data market, the fastest growing slice of AI data

$17B
world spend on collecting and labeling data by 2030, the budget we replace

Sources: Gartner (synthetic overtakes real by 2030; 60% of AI development data synthetic by 2024; 75% by 2026); Grand View Research (data collection and labeling market $17.1B by 2030 at ~28% CAGR; synthetic data generation ~35% CAGR, corroborated by MarketsandMarkets, IndustryARC and Next Move); Epoch AI and related studies (usable human data exhausted 2026 to 2032); public disclosures from NVIDIA, Tesla, Waymo, Meta and Google on synthetic and simulation training.

One engine. Eight vision markets, each in the billions.

RETAIL
$18B 2025
Shelf analytics, planograms, loss prevention. Amazon Just Walk Out.

MANUFACTURING
$20B 2025
Defect inspection at 95 to 99%. Generate rare defects you cannot collect. Foxconn, TSMC, Bosch.

AUTOMOTIVE
$19B 2025
Edge-case perception. Rare road scenes teams wait years to see. Waymo, Mobileye.

AGRICULTURE
$18B 2025
Crop, weed and disease. John Deere See & Spray cut herbicide ~50%. New classes in minutes.

ROBOTICS
$4.4B 2025
On-demand perception. A fast detector for each new object. Figure, Boston Dynamics.

LOGISTICS
$22B 2025
Warehouse safety. Forklift and worker detection. 25,110 US forklift injuries a year. Voxel AI ($64M).

AVIATION
$12B 2025
Runway debris, apron monitoring, aircraft inspection, bird and drone strikes.

MILITARY
$9.3B 2025
Air-gapped ISR. Detectors for imagery you cannot collect, fully on-prem. Anduril, Palantir Maven.

BEYOND EXPENSIVE TO COLLECT, INTO IMPOSSIBLE

Some things have **no dataset at all**: satellites and debris in orbit, rare defects, rare targets. You cannot photograph what you cannot reach, so we **generate the images and the labels**. Space domain awareness alone: **$1.9B** in 2025, almost entirely synthetic.

Sources (current market size, 2025 estimates): Grand View Research (retail CV, machine vision, AI in military, space domain awareness); MarketsandMarkets (agricultural robots, embodied AI); LogisticsIQ (warehouse automation); Mordor (inspection drones); ResearchAndMarkets (automotive AI). Also NSC (forklift injuries), Voxel AI, John Deere, Anduril, Palantir. Named firms operate in these categories; adoption paths are illustrative.

Priced like infrastructure, not labor.

PLAN	HOW IT IS PRICED	WHO IT LANDS
Developer	Free and open. Self-host, your own GPU and LLM. Export YOLO, COCO, ONNX.	Developers, researchers
Team	Per GPU-minute on managed MI300X. Every run quoted first. Shared projects and registry.	Most popular. A 500 image run costs about a lunch.
Enterprise	Dedicated or on-prem MI300X. SSO, isolation, SLAs, zero external calls.	Regulated, IP sensitive, defense

BUYER	ANNUAL SPEND
SMB or startup	$3k to $25k
Mid market	$50k to $250k
Enterprise	$300k to $1M plus
Gov or defense	$500k to $5M plus

> One GPU hour replaces about **$1,000** of human labeling. Margins track cloud GPU economics (70 to 85%), not human-in-the-loop services (30 to 40%).

The in-app pricing engine meters per GPU minute and quotes every run. Annual spend figures are reasoned estimates for planning, not quotes.

Everyone else needs your data, or your clicks.

	SCALE AI · LABELBOX	ROBOFLOW · ULTRALYTICS	AIONVIS
Who creates the data	You collect it	You collect it	**We generate it**
Who draws the boxes	Humans, by hand	Humans, assisted	**The agent swarm**
Humans in the loop	Many	Fewer, still required	**Zero**
Label trust layer	Human QA	Human review	**Self-verifying Critic**
Time to a new detector	Weeks	Days	**Minutes**

BUILT TODAY **22 architectures** · **5 task types** detect / segment / OBB / pose / classify · export to **.pt, ONNX, TorchScript, OpenVINO** · **YOLO / COCO / VOC / CSV** datasets

> Meta paid ~**$29B** for Scale AI, a business built on human labor in the loop. Our self-correcting **Critic is the moat**: everyone can generate, almost no one can trust it.

Sources: TechCrunch, CNBC (Scale AI, Meta, June 2025); Ultralytics; project run history measured in the repo.

THE CLOSE

Not another annotation tool. We give machines their eyes.

One sentence becomes a trained, verified detection model on a single AMD MI300X. Humans stop labeling. Agents start requesting their own vision.

REMOVES

The labeling tax. No data collection, no per box labor, no paying over and over.

SHIPS

The trust layer. A self-correcting Critic that makes synthetic data trainable.

GIVES

Eyes to agents on demand, through a headless API, resident in one MI300X.

Stop labeling. Start describing.